Fine-tuning ESM3 to design MerTK binders for macrophage efferocytosis

I want to improve macrophage clearance of dying cells in skin wounds, a process called efferocytosis. Impaired MerTK signaling has been linked to defective clearance and delayed wound repair.[1] I will develop an ESM3 fine-tuning approach for binder design, with MerTK as the lead target and AXL and TYRO3 as comparisons. Seitz et al. found a stronger requirement for MerTK in the mouse macrophages they tested,[2] and a human macrophage study also supports its role.[3]

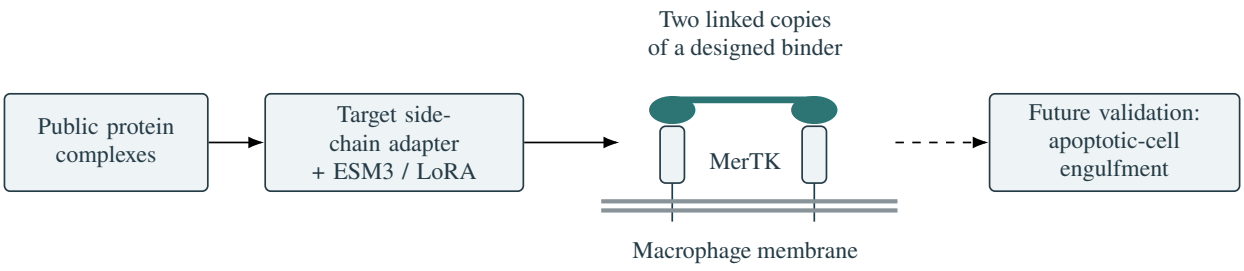
The protein would be a small binder to the extracellular region of MerTK, selected to leave room for its natural ligands, GAS6 and PROS1. I would link two copies to explore whether bringing MerTK receptors together improves clearance. Engineered antibodies provide a precedent for MerTK-mediated phagocytosis,[4] but activation alone can be misleading: one agonistic antibody impaired efferocytosis in arthritis models.[5] Success would require selective binding and increased engulfment while preserving natural ligand function.

ESM3 already uses sequence and backbone structure.[6] I propose adding a small geometric adapter to its open 1.4-billion-parameter model that describes nearby target side-chain atoms relative to each redesigned binder residue. I will train this adapter together with low-rank adaptation (LoRA), keeping the original ESM3 weights frozen. Building on earlier structural-adapter methods,[7] my hypothesis is that explicit side-chain context will improve interface sequence design with fewer training examples than LoRA alone. Training will reconstruct masked interface residues in public protein complexes, with their native side chains removed to prevent information leakage.

I will separate related protein families across training and evaluation sets and reserve the MerTK/AXL/TYRO3 family for the application. Unmodified ESM3 will serve as the baseline. Ablation studies will remove either the geometric adapter or LoRA to isolate each component's contribution. ProteinMPNN[8] and a small decoder trained on the same atomic inputs will provide additional benchmarks, with identical redesign masks across all models. Learning curves will measure interface sequence recovery, sequence diversity, and predicted complex consistency. I will use BindCraft[9] to generate candidate binder backbones and initial sequences in complex with MerTK. I will then use the fine-tuned ESM3 model to redesign interface residues while keeping the backbones fixed. Selection will consider glycan and membrane clashes, natural-ligand access, and predicted interactions with AXL and TYRO3. I will use open data and run computations on UCLA's Hoffman2 cluster and two NVIDIA RTX 6000 Ada graphics cards available in my lab. The project will produce a computational benchmark and ranked binder candidates.

The immediate goal is to establish whether the adapter improves interface design on held-out protein families. The main uncertainty is whether predicted binding translates into improved macrophage clearance.

As a future extension, I would test promising candidates in human macrophages using labeled apoptotic cells to distinguish internalization from surface binding. Monomeric and bivalent binders, a nonbinding mutant, and untreated cells would be tested with and without PROS1. MerTK-deficient cells would test receptor dependence, with receptor phosphorylation and inflammatory cytokines as supporting readouts.



Proposed workflow and mechanism. The dashed arrow indicates experimental validation proposed as a future extension.

References

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